A MORPHISM OF ALGEBRAIC STACKS IS UNRAMIFIED EXACTLY WHEN IT IS LOCALLY OF FINITE TYPE WITH ETALE DIAGONAL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the Stacks Project keeps one single shared preamble for every chapter; this card takes it directly from the staged sources/stacks-algebraic-stacks-morphisms/preamble.tex as three verbatim spans (lines 8-12, 14-18 and 154-217), so no derived file was written and nothing was added or redefined. The wrapper class and the base packages are supplied by the assembler.

Editorial note. Bibliography provenance: neither the counted statement nor its proof contains a citation (this lane rescans the whole span before writing the card), so the card carries no bibliography and no entry was reconstructed.

Editorial note. Selection provenance: this card was opened under the file-based control contract (CONTROL.md), with the mechanical pre-card gate run on its statement and proof interval before the card was written.

Allowed standard prerequisites (editorial).

- The description of a DM morphism through a surjective etale presentation of the fibre product over a smooth presentation of the target. Invoked in both directions of the counted proof (source lines 8460-8463 and 8508-8513) with a pointer to the author lemma lemma-DM of this chapter. Provides the presentation on which the property is read off.
- The criteria for checking a property of morphisms of algebraic stacks after a smooth covering base change and after precomposing with a surjective etale morphism, and the characterisation of unramified morphisms of algebraic spaces by their diagonal. Invoked inside the counted proof (source lines 8481-8484, 8494-8505, 8530-8535) as pointers to results of this and other chapters. Moves the etaleness of the diagonal up and down the presentation and turns the etaleness of a diagonal of an algebraic space into unramifiedness.

Source reference. Original lemma lemma-DM: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 7800–7800. Navigation pointer only: the counted text refers to the Stacks item lemma-DM, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original lemma lemma-etale-permanence: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 8527–8527. Navigation pointer only: the counted text refers to the Stacks item lemma-etale-permanence, which belongs to another chapter of the project and is not part of this card. The

Date: stacks-project:stacks-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference spaces-descent-lemma-etale-etale-local-source: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 8505–8505. Navigation pointer only: the counted text refers to the Stacks item spaces-descent-lemma-etale-etale-local-source, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference spaces-morphisms-lemma-diagonal-unramified-morphism: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 4660–4660. Navigation pointer only: the counted text refers to the Stacks item spaces-morphisms-lemma-diagonal-unramified-morphism, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference spaces-morphisms-lemma-etale-permanence: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 4653–4653. Navigation pointer only: the counted text refers to the Stacks item spaces-morphisms-lemma-etale-permanence, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference spaces-morphisms-lemma-etale-universally-injective-open: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 4656–4656. Navigation pointer only: the counted text refers to the Stacks item spaces-morphisms-lemma-etale-universally-injective-open, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference spaces-morphisms-lemma-unramified-locally-finite-type: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 8467–8467. Navigation pointer only: the counted text refers to the Stacks item spaces-morphisms-lemma-unramified-locally-finite-type, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference stacks-properties-lemma-check-property-after-precomposing: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 4242–4242. Navigation pointer only: the counted text refers to the Stacks item stacks-properties-lemma-check-property-after-precomposing, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference stacks-properties-lemma-check-property-weak-covering: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 745–745. Navigation pointer only: the counted text refers to the Stacks item stacks-properties-lemma-check-property-weak-covering, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's

own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $f : \mathcal{X} \to \mathcal{Y}$ be a morphism of algebraic stacks. The following are equivalent*

(1) *f is unramified, and*
(2) *f is locally of finite type and its diagonal is étale.*

Proof. Assume f is unramified. Then f is DM hence we can choose algebraic spaces U, V, a smooth surjective morphism $V \to \mathcal{Y}$ and a surjective étale morphism $U \to \mathcal{X} \times_{\mathcal{Y}} V$ (Lemma lemma-DM). Since f is unramified the induced morphism $U \to V$ is unramified. Thus $U \to V$ is locally of finite type (Morphisms of Spaces, Lemma spaces-morphisms-lemma-unramified-locally-finite-type) and we conclude that f is locally of finite type. The diagonal $\Delta : \mathcal{X} \to \mathcal{X} \times_{\mathcal{Y}} \mathcal{X}$ is a morphism of algebraic stacks over $\mathcal{Y}$. The base change of Δ by the surjective smooth morphism $V \to \mathcal{Y}$ is the diagonal of the base change of f, i.e., of $\mathcal{X}_V = \mathcal{X} \times_{\mathcal{Y}} V \to V$. In other words, the diagram

$$\begin{array}{ccc} \mathcal{X}_V & \longrightarrow & \mathcal{X}_V \times_V \mathcal{X}_V \\ \downarrow & & \downarrow \\ \mathcal{X} & \longrightarrow & \mathcal{X} \times_{\mathcal{Y}} \mathcal{X} \end{array}$$

is cartesian. Since the right vertical arrow is surjective and smooth it suffices to show that the top horizontal arrow is étale by Properties of Stacks, Lemma stacks-properties-lemma-check-property-weak-covering. Consider the commutative diagram

$$\begin{array}{ccc} U & \longrightarrow & U \times_V U \\ \downarrow & & \downarrow \\ \mathcal{X}_V & \longrightarrow & \mathcal{X}_V \times_V \mathcal{X}_V \end{array}$$

All arrows are representable by algebraic spaces, the vertical arrows are étale, the left one is surjective, and the top horizontal arrow is an open immersion by Morphisms of Spaces, Lemma spaces-morphisms-lemma-diagonal-unramified-morphism. This implies what we want: first we see that $U \to \mathcal{X}_V \times_V \mathcal{X}_V$ is étale as a composition of étale morphisms, and then we can use Properties of Stacks, Lemma stacks-properties-lemma-check-property-after-precomposing to see that $\mathcal{X}_V \to \mathcal{X}_V \times_V \mathcal{X}_V$ is étale because being étale (for morphisms of algebraic spaces) is local on the source in the étale topology (Descent on Spaces, Lemma spaces-descent-lemma-etale-etale-local-source).

Assume f is locally of finite type and that its diagonal is étale. Then f is DM by definition (as étale morphisms of algebraic spaces are unramified). As above this means we can choose algebraic spaces U, V, a smooth surjective morphism $V \to \mathcal{Y}$ and a surjective étale morphism $U \to \mathcal{X} \times_{\mathcal{Y}} V$ (Lemma lemma-DM). To finish the proof we have to show that $U \to V$ is unramified. We already know that $U \to V$ is locally of finite type. Arguing as above we find a commutative diagram

$$\begin{array}{ccc} U & \longrightarrow & U \times_V U \\ \downarrow & & \downarrow \\ \mathcal{X}_V & \longrightarrow & \mathcal{X}_V \times_V \mathcal{X}_V \end{array}$$

where all arrows are representable by algebraic spaces, the vertical arrows are étale, and the lower horizontal one is étale as a base change of Δ. It follows that $U \to$

$U \times_V U$ is étale for example by Lemma lemma-etale-permanence[1]. Thus $U \to U \times_V U$ is an étale monomorphism hence an open immersion (Morphisms of Spaces, Lemma spaces-morphisms-lemma-etale-universally-injective-open). Then $U \to V$ is unramified by Morphisms of Spaces, Lemma spaces-morphisms-lemma-diagonal-unramified-morphism. □

[1]It is quite easy to deduce this directly from Morphisms of Spaces, Lemma spaces-morphisms-lemma-etale-permanence.